

Name: _____

Date: _____

VECTORS, (VECTOR GEOMETRY PROOFS)

LO: I can use vectors to prove geometric properties such as collinearity and parallelism.

Vector proofs

In vector geometry, we prove properties by showing vectors are **parallel** (scalar multiples) or that points are **collinear** (lie on the same line).

Key technique: Express vectors in terms of given vectors a and b . If one vector is a scalar multiple of another, they are parallel.

Proof techniques

●	If $AB = kAC$, then A, B, C are collinear	same direction, common point
●	If $PQ = kRS$, then PQ is parallel to RS	scalar multiple
●	Midpoint M: $OM = \frac{1}{2}(OA + OB)$	midpoint formula
●	Same magnitude means parallel	direction must match

Key idea

Express unknown vectors using known ones. Show scalar multiples for parallel/collinear.

$$AB = \overrightarrow{OB - OA} = \mathbf{b - a}$$

Route from A to B via the origin

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - expressing vectors

$OA = a$, $OB = b$. Find AB.

$$\begin{aligned} AB &= AO + OB \\ &= -a + b \\ &= b - a \end{aligned}$$

$$AB = b - a$$

Go backwards along a , then forwards along b .

Example 2 - midpoint

M is the midpoint of AB. $OA = a$, $OB = b$. Find OM.

$$\begin{aligned} OM &= OA + AM \\ AM &= \frac{1}{2}AB = \frac{1}{2}(b - a) \\ OM &= a + \frac{1}{2}(b - a) = \frac{1}{2}(a + b) \end{aligned}$$

$$OM = \frac{1}{2}(a + b)$$

The midpoint formula in vectors.

Example 3 - proving parallel

$OA = a$, $OB = b$, $OC = 3a$, $OD = 3b$. Show AB is parallel to CD.

$$\begin{aligned} AB &= b - a \\ CD &= OD - OC = 3b - 3a = 3(b - a) \\ CD &= 3 \times AB, \text{ so parallel} \end{aligned}$$

$$CD = 3AB, \text{ so AB parallel to CD}$$

CD is a scalar multiple of AB.

Example 4 - proving collinear

$OA = 2a + b$, $OB = 5a + 4b$, $OC = 8a + 7b$. Prove A, B, C collinear.

$$\begin{aligned} AB &= OB - OA = 3a + 3b \\ AC &= OC - OA = 6a + 6b = 2(3a + 3b) \\ AC &= 2AB, \text{ common point A, so collinear} \end{aligned}$$

$$AC = 2AB, \text{ collinear}$$

AC is a scalar multiple of AB and they share point A.

YOUR TURN – PRACTICE (deliberate practice)

1. Expressing vectors

OA = a, OB = b. Express in terms of a and b.

- a) AB
- b) BA
- c) Midpoint M of AB: OM
- d) Point P where OP = 2a + b: AP

2. Scalar multiples

State if the vectors are parallel.

- a) $2a + 4b$ and $a + 2b$
- b) $3a - b$ and $6a - 2b$
- c) $a + 3b$ and $2a + 3b$
- d) $4a - 2b$ and $-2a + b$

3. Finding ratios

Find the ratio AB:AC.

- a) AB = 2a, AC = 6a
- b) AB = b - a, AC = 3b - 3a
- c) AB = 4a + 2b, AC = 8a + 4b
- d) AB = a + b, AC = 3a + 3b

4. Midpoints and ratios

M is midpoint of AB. OA = a, OB = b.

- a) Find AM
- b) Find OM
- c) P divides AB in ratio 1:2. Find OP.
- d) Q divides AB in ratio 2:1. Find OQ.

5. Proof problems

Solve each.

- a) OA=a, OB=b, OC=a+b. What shape is OACB?
- b) AB = 3a+6b, CD = a+2b. Relationship?
- c) P = a, Q = 4a. Are O, P, Q collinear?
- d) AB = 2i+3j, AC = 4i+6j. Collinear?
- e) Show ABCD is a parallelogram if AB = DC

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $AB = OB - OA = b - a$ ☐

Correct answer: _____

Reason: _____

b) $AB = BA$ ☐

Correct answer: _____

Reason: _____

c) Parallel vectors are scalar multiples of each other ☐

Correct answer: _____

Reason: _____

d) Collinear points must share a common vector direction ☐

Correct answer: _____

Reason: _____

e) If $|AB| = |CD|$ then $AB = CD$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) In triangle OAB, OA = a and OB = b. P is the midpoint of OA. Q is the point on AB such that AQ:QB = 1:2. Express PQ in terms of a and b.

Expression: _____

Simplified: _____

b) OABC is a parallelogram with OA = a and OC = c. M is the midpoint of AB. Prove that OM and AC intersect at a point that divides both in ratio 2:1.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

- Have I shown all my working clearly?
- Did I check my answer using a different method?
- Does my answer look reasonable?

